

Question

1. A highly toxic substance is obtained by the fusion reaction of calcium cyanamide, coke, and sodium carbonate, followed by water leaching.
2. When realgar reacts with an excess of concentrated nitric acid, a pale yellow precipitate is obtained.
3. Cobalt(III) nitrate undergoes thermal decomposition to yield cobalt(II,III) oxide.
4. Rhenium hexafluoride undergoes disproportionation in water.
5. $\text{Ga}_2(\text{OH})_2\text{Cl}_2 \cdot 2\text{H}_2\text{O}$ dissolves in NaOH solution, releasing gas.

Let the sums of the simplest integer ratios of the balanced chemical equations for the above five reaction processes be denoted as a, b, c, d, and e, respectively. Then, which of the following statements is correct?

A. There does not exist a number greater than 50.

B. There are 2 numbers greater than 30.

C. There are 2 numbers less than 10.

D. There are 3 odd numbers

E. There are 2 prime numbers

F. All of the above statements are incorrect

Answer

Correct Answer: **E**

Detailed Explanation

For Reaction 1, the structural formula of calcium cyanamide is CaCN_2 . Considering the "highly toxic substance" mentioned in the question stem, it can only be NaCN , and the destination of Ca is CaCO_3 . The equation is: $\text{CaCN}_2 + \text{Na}_2\text{CO}_3 + \text{C} = \text{CaCO}_3 + 2\text{NaCN}$, with a sum of 6.

CHECKPOINT

1 PTS

Reaction 1: $\text{CaCN}_2 + \text{Na}_2\text{CO}_3 + \text{C} = \text{CaCO}_3 + 2\text{NaCN}$

CHECKPOINT

0.5 PTS

Sum of coefficients for Reaction 1 is 6

For Reaction 2, the "pale yellow precipitate" mentioned in the question stem should be S . Since concentrated nitric acid is added, the product should be the highest oxidation state of As , and the concentrated nitric acid is reduced to NO_2 . The equation is: $\text{As}_4\text{S}_4 + 20\text{HNO}_3 = 4\text{H}_3\text{AsO}_4 + 4\text{S} + 20\text{NO}_2 + 4\text{H}_2\text{O}$, with a sum of 53.

CHECKPOINT

1 PTS

Reaction 2: $\text{As}_4\text{S}_4 + 20\text{HNO}_3 = 4\text{H}_3\text{AsO}_4 + 4\text{S} + 20\text{NO}_2 + 4\text{H}_2\text{O}$

CHECKPOINT

0.5 PTS

Sum of coefficients for Reaction 2 is 53

For Reaction 3, it can be observed that the decomposition involves redox, so N should exist as NO_2 . The equation is: $6\text{Co}(\text{NO}_3)_3 = 2\text{Co}_3\text{O}_4 + 18\text{NO}_2 + 5\text{O}_2$, with a sum of 31.

CHECKPOINT

1 PTS

Reaction 3: $6\text{Co}(\text{NO}_3)_3 = 2\text{Co}_3\text{O}_4 + 18\text{NO}_2 + 5\text{O}_2$

CHECKPOINT

0.5 PTS

Sum of coefficients for Reaction 3 is 31

For Reaction 4, disproportionation generally forms adjacent oxidation states. Based on elemental properties, +6-valent Re disproportionates to +7 and +4. The equation is: $3\text{ReF}_6 + 10\text{H}_2\text{O} = 2\text{HReO}_4 + \text{ReO}_2 + 18\text{HF}$, with a sum of 34.

CHECKPOINT

1 PTS

Reaction 4: $3\text{ReF}_6 + 10\text{H}_2\text{O} = 2\text{HReO}_4 + \text{ReO}_2 + 18\text{HF}$

CHECKPOINT

0.5 PTS

Sum of coefficients for Reaction 4 is 34

For Reaction 5, +2-valent Ga is strongly reducing and forms H_2 . The equation is: $Ga_2(OH)_2Cl_2 \cdot 2H_2O + 4NaOH = 2NaGa(OH)_4 + 2NaCl + H_2$, with a sum of 10.

CHECKPOINT

1 PTS

Reaction 5: $Ga_2(OH)_2Cl_2 \cdot 2H_2O + 4NaOH = 2NaGa(OH)_4 + 2NaCl + H_2$

CHECKPOINT

0.5 PTS

Sum of coefficients for Reaction 5 is 10

The sum of coefficients for Reaction 2 is greater than 50, so A is incorrect. The sums for Reactions 2, 3, and 4 are greater than 30, so B is incorrect. Only Reaction 1 has a sum of coefficients less than 10, so C is incorrect. Only Reactions 2 and 3 have sums that are odd, so D is incorrect. The sums for Reactions 2 and 3 are prime numbers, so E is correct.